Magnetism and electromagnetism

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 Units [1]

Answer: B

Explanation

The watt (W) is the unit of power. Current is measured in amperes (A), voltage in volts (V) and energy in joules (J), so A, C and D are wrong. D is a common confusion: power is the rate of transfer of energy, not the energy itself.

2 · Part 1

Paper 1 & 2 Magnetism [1]

Answer: D

Worked solution

Only repulsion proves that an object is a magnet. Attraction to a north pole happens either because the bar is a magnet with a south pole at that end, or because magnetism is induced in an unmagnetised magnetic material.

3 · Part 1

Paper 1 & 2 Magnetism [1]

Answer: A

Explanation

A magnetically soft material (such as iron) is magnetised easily and loses its magnetism when the magnetising field is removed. C describes a magnetically hard material (such as steel), which is harder to magnetise but keeps its magnetism. B and D do not describe soft materials: D is a non-magnetic material, and B combines the "easy to magnetise" property of soft materials with the "keeps its magnetism" property of hard materials.

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4 · Part 1 Paper 1 & 2 Electromagnetism [1]

Answer: C

Worked solution

Hold the left hand so the first finger points down and the second finger points east. The thumb then points north. The force is always perpendicular to both the current and the field, so it cannot be up, down, east or west here.

5 · Part 1 Paper 2 only Electromagnetism [1]

Answer: C

Explanation

A charged particle moving in a magnetic field experiences a force as long as its motion is not parallel to the field. Moving parallel to the field lines gives no force. In A, B and D the motion is not parallel to the field, so there is a force (largest in A, when the motion is perpendicular).

6 · Part 1 Paper 1 & 2 Electromagnetism [1]

Answer: A

Explanation

Increasing the current increases the size of the force. Reversing the field reverses the direction of the force. So the force is larger and reversed. B ignores the reversal of the field; C ignores the increase in current; D wrongly thinks a larger current gives a smaller force.

7 · Part 1 Paper 1 & 2 Electromagnetic induction [1]

Answer: D

Explanation

A stronger magnet increases the induced voltage. Fewer turns (A) and slower movement (B) reduce the induced voltage. Holding the magnet still (C) gives no change of field through the coil, so no voltage is induced.

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8 · Part 1 Paper 2 only Electromagnetic induction [1]

Answer: B

Explanation

A step-up transformer increases the voltage, so its secondary coil has more turns than its primary coil. A is wrong: a steady d.c. gives an unchanging magnetic field in the core, so no voltage is induced in the secondary. C is wrong: the core is iron, and the coils are linked only by the magnetic field, not by a current through the core. D is wrong: a transformer cannot increase power; at best (100% efficient) output power equals input power.

9 · Part 1 Paper 1 & 2 Magnetism [5]

Mark scheme

(a) **[1]**

- steel (cans) (1)

(b) **[2]**

- the first can is in the magnetic field of the magnet, so magnetism is induced in it OR it becomes a (temporary) magnet (1)
- (so) it attracts the second can, which is a magnetic material (1)

ALLOW steel is a magnetic material so the first can is magnetised by the magnet for the first mark

(c)(i) **[1]**

- they repel OR push apart (1)

(c)(ii) **[1]**

- they attract OR pull together (1)

Worked solution

(a) Steel contains iron, so it is a magnetic material and is attracted to a magnet. Aluminium is not a magnetic material.

(b) When a magnetic material is placed in a magnetic field, magnetism is induced in it. The first can becomes a temporary magnet while it is touching the magnet, so it can attract another piece of magnetic material: the second can.

(c) Like poles repel and unlike poles attract. Two north poles push apart; a north pole and a south pole pull together.

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10 · Part 1 Paper 1 & 2 Magnetism [6]

Mark scheme

(a) **[1]**

- R (1)

(b) **[2]**

- Q (1)
- (because) it still holds (8) paper clips after the magnet is removed OR it keeps (most of) its magnetism (1)
- The reason mark depends on choosing Q.

(c) **[2]**

- P (1)
- it becomes strongly magnetised while the magnet is held against it (picks up the most clips) AND loses all its magnetism when the magnet is removed (holds 0 clips), so the pins drop (1)

> **ALLOW** P is magnetically soft, so the pins are released when the magnet is removed for the second mark

- The explanation mark depends on choosing P.

(d) **[1]**

- any one from: same magnet OR same type or size of paper clip OR same time waited (one minute) OR same end held against the magnet OR same way of holding the rod against the magnet (1)

> **IGNORE** same rod size (excluded by the question)

Worked solution

(a) Rod R holds no paper clips even while it touches the magnet, so no magnetism is induced in it: it is non-magnetic.

(b) A magnetically hard material is hard to magnetise but keeps its magnetism. Rod Q still holds 8 clips a minute after the magnet is removed, so it has kept its magnetism.

(c) The tool must pick up pins only while the magnet is held on it, then release them. Rod P picks up 12 clips while touching the magnet but holds none once the magnet is removed. It is magnetically soft, so the pins drop as soon as the magnet is taken away. Rod Q would keep holding the pins.

(d) For a fair test, only the material should change. Keeping the same magnet, the same paper clips and the same waiting time are control variables.

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11 · Part 1 Paper 1 & 2 Magnetism [5]

Mark scheme

(a) **[1]**

- a line showing the direction of the (magnetic) force on a north pole placed in the field (1)

> **ALLOW** a line showing the direction in which a plotting compass needle points OR the direction of the magnetic field at each point

(b) **[1]**

- from the north pole to the south pole (1)

(c) **[2]**

- if two lines crossed, the field would have two different directions at that point (1)
- but a north pole (or compass needle) at a point can only be pushed (or point) in one direction OR the field has only one direction at any point (1)

(d) **[1]**

- the magnetic field is strong(er) there (1)

Worked solution

A magnetic field line shows the direction of the force that a north pole would feel at each point, which is the direction a plotting compass needle points. Outside a magnet the lines go from the north pole to the south pole. At any point the field has only one direction, so lines cannot cross: a crossing would mean two directions at once. The closer together the lines, the stronger the field, which is why lines bunch together near the poles.

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12 · Part 1 Paper 1 & 2 Magnetism [6]

Mark scheme

(a) **[4]**

Indicative points, any four from:

- place the magnet on the paper and draw around it (1)
- place the plotting compass near one pole (e.g. the north pole) and mark a dot at the end the compass needle points to (1)
- move the compass so that the other end of its needle (its tail) is over the dot and mark a new dot where the needle now points (1)
- repeat until the line reaches the other pole or the edge of the paper (1)
- join the dots with a smooth line and add an arrow showing the direction the needle points (1)
- repeat from several different starting points around the magnet (1)

(b) **[1]**

- any one from: another magnet OR a magnetic material (e.g. a steel bench, iron object) nearby OR the Earth's magnetic field affects the compass OR dots marked inaccurately (1)

(c) **[1]**

- the north-seeking end of the compass needle points away from the north pole of the magnet OR the south end of the compass needle points towards the north pole of the magnet (1)

ALLOW the north pole is the end that the north end of the compass needle is repelled from

Worked solution

(a) Draw around the magnet on the paper. Put the compass near a pole and mark a dot where the needle's north end points. Move the compass on so its tail is at the dot, and mark the new position of the needle's north end. Keep going, step by step, until you reach the other pole. Join the dots with a smooth curve and put an arrow on it in the direction the needle points. Start again from other points to build up the whole pattern.

(b) The compass responds to every magnetic field near it, so a nearby steel object, another magnet or the Earth's field can distort the pattern.

(c) Field lines leave the north pole, and the compass needle's north end points along them, so it points away from the magnet's north pole.

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13 · Part 1 Paper 1 & 2 Magnetism [6]

Mark scheme

(a) **[1]**

- so the filings can move and line up with the magnetic field OR to help the filings align along the field lines (1)

(b) **[2]**

- the field lines from each north pole curve away sideways OR spread out away from the gap OR do not join across the gap (1)
- (there is) a point (or region) midway between the poles with no (pattern of) filings OR a neutral point (1)

(c) **[2]**

- the two magnets produce fields of equal size (strength) at that point (1)
- (that are) in opposite directions, so they cancel OR the resultant field is zero (1)

(d) **[1]**

- iron filings do not show the direction of the field (1)

ALLOW filings are difficult to see where the field is weak OR the pattern is lost when the card is moved

Worked solution

(a) Tapping the card lets the filings move freely so they line up along the field lines.

(b) Field lines leave a north pole. With two north poles facing each other, the lines from each pole are pushed sideways and curve away; none join across the gap. Midway there is a neutral point with no field.

(c) The magnets are identical and the point is the same distance from each, so their fields have the same strength there. The field from each points away from its own north pole, so the two fields point in opposite directions and cancel.

(d) Iron filings show the shape of the pattern but not which way the field points; a plotting compass shows the direction.

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14 · Part 1 Paper 1 & 2 Magnetism [5]

Mark scheme

(a) **[2]**

- opposite (unlike) poles facing each other OR the north face of one magnet facing the south face of the other (1)
- (flat faces) parallel to each other AND close together OR with a small gap between them (1)

(b) **[2]**

- straight and parallel (1)
- equally spaced (1)

> **ALLOW** all pointing in the same direction (from north to south) for one mark if neither of the above is given

(c) **[1]**

- the field (strength) is the same everywhere in the gap, so the force on the wire does not depend on where the wire is placed (1)

> **ALLOW** so that the field is the same along the whole length of the wire in the field OR so that it is a fair test

Worked solution

Place the two flat magnets with a north face facing a south face, parallel and close together. Between them the field lines are straight, parallel and equally spaced, going from the north face to the south face: the field has the same strength and direction everywhere. This is useful because the force on the wire then depends only on what the teacher changes (such as the current), not on exactly where the wire is in the gap.

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15 · Part 1 Paper 1 & 2 Electromagnetism [6]

Mark scheme

(a) **[2]**

- the current in the rod produces a magnetic field around it (1)
- this field interacts with the field of the magnet (producing a force) (1)

> **ALLOW** a current-carrying conductor at right angles to a magnetic field experiences a force for the second mark only

(b) **[1]**

- towards the student (1)

(c) **[2]**

- reverse the (direction of the) current (1)
- reverse the (direction of the) magnetic field OR turn the magnet upside down OR swap the poles (1)

> **REJECT** both changes made together for either mark

(d) **[1]**

- any one from: increase the current OR use a stronger magnet (1)

> **ALLOW** increase the voltage of the supply

Worked solution

(a) A current in a conductor produces a magnetic field around it. This field and the magnet's field interact, so there is a force on the rod.

(b) Left-hand rule: point the first finger along the field (up) and the second finger along the current (to the right). The thumb points towards the student, so the rod rolls towards the student.

(c) Reversing either the current or the field reverses the force. Reversing both at once would leave the force in the original direction.

(d) A larger current or a stronger magnetic field gives a larger force in the same direction.

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16 · Part 1 Paper 1 & 2 Electromagnetism [8]

Mark scheme

(a) **[4]**

Indicative points, any four from:

- there is a current in the coil (when the switch is closed) (1)
- (the long) sides of the coil are at right angles to the magnetic field, so there is a force on each side (1)
- the current in one side is in the opposite direction to the current in the other side (1)
- so the forces on the two sides are in opposite directions OR one side is pushed up and the other is pushed down (1)
- these forces produce a turning effect on the coil, so it rotates (on the axle) (1)

ALLOW the magnetic field of the current in the coil interacts with the field of the magnets for the second mark

(b) **[2]**

- it reverses the (direction of the) current in the coil every half turn (1)
- so the forces keep turning the coil in the same direction OR so the coil keeps rotating continuously in one direction (1)

(c) **[2]**

- any two from: increase the current OR increase the voltage of the battery OR use a stronger magnetic field (magnets) OR increase the number of turns on the coil (1) (1)

Worked solution

(a) When the switch is closed there is a current around the coil. Each long side is a current-carrying wire at right angles to the magnetic field, so each side feels a force. Because the current goes along one side and comes back along the other, the currents are in opposite directions, so by the left-hand rule one side is pushed up and the other down. These opposite forces turn the coil about the axle.

(b) After half a turn the sides have swapped places. Without the commutator, the forces would now turn the coil back. The split-ring commutator reverses the current in the coil every half turn, so the force on whichever side is on the left is always in the same direction and the coil keeps rotating the same way.

(c) A bigger current, stronger magnets or more turns all give a bigger force on the coil, so it turns faster.

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17 · Part 1 Paper 1 & 2 Electromagnetism [6]

Mark scheme

(a) **[3]**

- the current in the coil is in a magnetic field, so there is a force on the coil (1)
- the alternating current keeps reversing direction, so the force (on the coil) keeps reversing direction (1)
- (so) the coil and cone are pushed backwards and forwards (vibrate) (1)

(b) **[1]**

- 250 Hz (1)

(c) **[2]**

- larger current gives a larger force on the coil (1)
- so the cone moves further backwards and forwards OR vibrates with a larger amplitude (1)

> **IGNORE** louder

Worked solution

(a) The coil carries a current in the magnet's field, so there is a force on it. The current is alternating, so it keeps changing direction, and the force on the coil keeps changing direction too. The coil, and the cone attached to it, are pushed out and pulled in again and again: the cone vibrates.

(b) The force reverses each time the current reverses, so the cone vibrates at the same frequency as the current: 250 Hz.

(c) The force on a current-carrying conductor in a magnetic field increases when the current increases. A larger force moves the cone further, so it vibrates with a larger amplitude, at the same frequency.

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18 · Part 1 Paper 2 only Electromagnetism [7]

Mark scheme

(a) **[3]**

- a coil of (insulated) wire (1)
- wrapped around an (soft) iron core (1)
- connected to a (d.c.) electricity supply OR battery (and a switch) (1)

ALLOW solenoid for coil

(b) **[2]**

- increase the current (1)
- increase the number of turns on the coil (1)

ALLOW increase the voltage of the supply as increasing the current

REJECT use an iron core (the electromagnet already has one)

(c) **[2]**

- iron is magnetically soft OR iron loses its magnetism when the current is switched off (1)
- so the door (steel plate) is released and the door closes OR steel would stay magnetised and keep holding the door open (1)

Worked solution

(a) An electromagnet is a coil of insulated wire wound around a soft iron core and connected to a power supply. A current in the coil produces a magnetic field; the iron core makes the field much stronger.

(b) A larger current or more turns of wire make the magnetic field stronger.

(c) For a fire door the magnet must let go as soon as the current is switched off. Iron is magnetically soft, so it loses its magnetism straight away and the door is released. Steel is magnetically hard and would stay magnetised, so it could keep holding the door open in a fire.

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19 · Part 1 Paper 2 only Electromagnetism [6]

Mark scheme

(a) **[2]**

- electrons are (negatively) charged (particles) (1)
- an electric current is a flow of charge OR a current is the rate of flow of charge, so moving electrons form a current (1)

(b) **[3]**

- (conventional) current is in the opposite direction to the electron flow, so the current is from right to left (1)
- left-hand rule applied with the field into the screen (away from the student) and the current to the left (1)
- force is downwards (1)

ALLOW full marks for downwards with reasoning that the force on negative electrons is opposite to the force on positive charges moving the same way

ALLOW a maximum of 1 mark (the second marking point) for upwards obtained by applying the left-hand rule correctly to a current from left to right

(c) **[1]**

- upwards (1)

ALLOW ECF: the opposite direction to the answer given in (b)

Worked solution

(a) Electrons carry negative charge. A current is a flow of charge, so a beam of moving electrons is a current, even without a wire.

(b) Conventional current is taken as the direction positive charge would flow, which is opposite to the electrons. The electrons move left to right, so the current is right to left. Using the left-hand rule, first finger into the screen (field) and second finger to the left (current): the thumb points downwards. The force on the electrons is downwards.

(c) Protons are positive, so for protons moving left to right the current is left to right. This reverses the current compared with (b), so the force reverses: upwards.

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20 · Part 1 Paper 1 & 2 Electromagnetic induction [7]

Mark scheme

(a)(i) **[1]**

- zero OR no reading OR the pointer stays at zero (1)

(a)(ii) **[1]**

- the pointer moves to the left (of zero) (by the same amount) (1)

ALLOW a reading of the same size in the opposite direction

(b)(i) **[2]**

- the induced voltage increases as the speed increases (1)
- the induced voltage is (directly) proportional to the speed OR doubling the speed doubles the voltage (1)

(b)(ii) **[1]**

- 3.0 mV (1)

ALLOW 3 mV

(c) **[2]**

- the wire is moving parallel to the magnetic field (lines) OR along the field lines (1)
- so the wire does not cut any field lines OR the field around the wire does not change, so no voltage is induced (1)

Worked solution

(a) A voltage is only induced while the wire moves through the field and cuts field lines. Held still, the reading is zero. Moving upwards instead of downwards reverses the induced voltage, so the pointer moves the same amount to the left.

(b) Each time the speed goes up by 0.10 m/s, the voltage goes up by 0.6 mV; doubling the speed from 0.10 m/s to 0.20 m/s doubles the voltage from 0.6 mV to 1.2 mV. The voltage is directly proportional to the speed. At 0.50 m/s: 0.6 mV × 5 = 3.0 mV.

(c) The field runs from the north pole to the south pole. Moving the wire in that direction moves it along the field lines, so it cuts none of them and no voltage is induced.

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21 · Part 1 Paper 1 & 2 Electromagnetic induction [8]

Mark scheme

(a) **[2]**

- the magnetic field through the coil changes (as the magnet rotates) (1)
- (a changing magnetic field through a coil) induces a voltage (in the coil) (1)

ALLOW the coil cuts magnetic field lines for the first mark

(b) **[2]**

- the north and south poles of the magnet pass the coil in turn OR the direction of the magnetic field through the coil reverses every half turn (1)
- so the direction of the induced voltage reverses (repeatedly) (1)

(c) **[2]**

- the (peak) voltage is larger (1)
- the voltage alternates more often OR the frequency (of the a.c.) is higher (1)

(d) **[2]**

- any two from: number of turns on the coil OR strength of the magnetic field (magnet) OR area of the coil (1) (1)

Worked solution

(a) As the magnet turns, the magnetic field passing through the coil keeps changing. A changing magnetic field through a coil induces a voltage in it.

(b) Each half turn, the north pole and then the south pole face the coil, so the field through the coil reverses. When the change of field reverses, the induced voltage reverses too, so the output is alternating.

(c) Faster rotation means the field changes faster, so the induced voltage is larger. It also means more turns of the magnet each second, so the voltage reverses more often: the frequency increases.

(d) The size of the induced voltage also depends on the number of turns on the coil, the strength of the magnetic field and the area of the coil.

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22 · Part 1 Paper 2 only Electromagnetic induction [7]

Mark scheme

(a) **[3]**

- two coils of (insulated) wire, called the primary and secondary coils (1)
- wound on a (soft) iron core (1)
- the coils are not electrically connected to each other OR the coils are linked only by the core (1)

ALLOW different numbers of turns on the two coils for the third mark

(b) **[3]**

- (the alternating voltage produces) an alternating current in the primary coil (1)
- (this produces) a changing (alternating) magnetic field in the iron core (1)
- the changing magnetic field passes through the secondary coil and induces an (alternating) voltage across it (1)

(c) **[1]**

- so that current flows through every turn of the coil, not straight across between neighbouring turns OR so that current does not pass into the (iron) core OR to stop a short circuit (1)

Worked solution

(a) A transformer has a primary coil and a secondary coil of insulated wire, wound on the same soft iron core. The coils are not joined by any wire; they are linked only through the core.

(b) The alternating voltage drives an alternating current in the primary coil. This current makes a magnetic field in the core that keeps changing size and direction. The core carries this changing field through the secondary coil, and a changing magnetic field through a coil induces a voltage. So an alternating voltage appears across the secondary coil.

(c) If the turns touched with bare wire, the current would take a short cut between neighbouring turns (or through the core) instead of going round every turn, and the transformer would not work properly.

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23 · Part 1 Paper 2 only Electromagnetic induction [7]

Mark scheme

(a) **[1]**

- step-down (1)

(b) **[3]**

- input (primary) voltage / output (secondary) voltage = primary turns / secondary turns OR 230 / 12 = 1840 / Ns (1)
- Ns = 1840 × 12 / 230 (1)
- 96 (turns) (1)
- Correct answer with no working scores 3 marks.

(c) **[3]**

- Vp × Ip = Vs × Is OR input power = output power (1)
- 230 × Ip = 12 × 2.3 OR Ip = 27.6 / 230 (1)
- 0.12 A (1)
- Correct answer with no working scores 3 marks.

Worked solution

(a) The output voltage (12 V) is lower than the input voltage (230 V), so it is a step-down transformer.

(b) Vp / Vs = Np / Ns, so 230 / 12 = 1840 / Ns. Rearranging: Ns = 1840 × 12 / 230 = 22 080 / 230 = 96 turns. Check: 1840 / 96 = 19.2 (to 3 s.f.), and 230 / 12 = 19.2 (to 3 s.f.), so the ratios match.

(c) For a 100% efficient transformer, input power = output power, so Vp × Ip = Vs × Is. Output power = 12 × 2.3 = 27.6 W. Ip = 27.6 / 230 = 0.12 A. Check: the voltage is stepped down by a factor of about 19, so the current is stepped up by the same factor: 0.12 × 19.2 ≈ 2.3 A.

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24 · Part 1 Paper 2 only Electromagnetic induction [6]

Mark scheme

(a) **[4]**

Indicative points, any four from:

- for the same power transmitted, a higher voltage means a smaller current (power = current × voltage) (1)
- a smaller current causes less heating of the cables (1)
- so less energy is wasted (to the surroundings as thermal energy) (1)
- so transmission is more efficient OR more of the energy reaches the consumers (1)
- (a smaller current means) thinner (cheaper, lighter) cables can be used (1)

(b) **[2]**

- very high voltages are (extremely) dangerous (in homes) (1)
- homes and appliances are designed to work at a much lower voltage OR the voltage must be reduced to about 230 V (1)

Worked solution

(a) Power = current × voltage. A power station sends out a fixed power, so if the voltage is made very high, the current in the cables is very small. Current heats the cables, wasting energy to the surroundings. A small current means much less heating, so much less energy is wasted and the transmission is more efficient. Smaller currents also allow thinner, cheaper cables to be used.

(b) A voltage of hundreds of thousands of volts would be deadly in a home and would damage appliances designed for about 230 V. Step-down transformers reduce the voltage to a safe, usable value before it reaches homes.

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25 · Part 1 Paper 2 only Electromagnetic induction [8]

Mark scheme

(a) **[3]**

- power = current × voltage OR Vs × Is = 160 MW (1)
- Is = 160 000 000 / 400 000 (1)
- 400 A (1)
- Correct answer with no working scores 3 marks.

(b) **[2]**

- Ip = 160 000 000 / 25 000 OR Vp × Ip = Vs × Is with 25 000 × Ip = 400 000 × 400 (1)
- 6400 A (1)

ALLOW ECF from (a) when using Vp × Ip = Vs × Is

(c) **[2]**

- 25 / 400 = 750 / Ns OR Ns = 750 × 400 / 25 (1)
- 12 000 (turns) (1)
- Correct answer with no working scores 2 marks.

(d) **[1]**

- heating of the coils OR heating of the (iron) core OR energy transferred to the surroundings as thermal energy (1)

ALLOW energy wasted as sound (hum)

Worked solution

(a) Power = current × voltage, with P = 160 MW = 160 000 000 W and V = 400 kV = 400 000 V. I = 160 000 000 / 400 000 = 400 A.

(b) The transformer is 100% efficient, so the input power is also 160 MW. Ip = 160 000 000 / 25 000 = 6400 A. Check with Vp × Ip = Vs × Is: 25 000 × 6400 = 160 000 000 and 400 000 × 400 = 160 000 000, which agree.

(c) Vp / Vs = Np / Ns, so 25 / 400 = 750 / Ns, giving Ns = 750 × 400 / 25 = 750 × 16 = 12 000 turns. The voltage goes up by a factor of 16, so the turns go up by a factor of 16 as well.

(d) In a real transformer some energy is wasted, mainly by heating of the coils and the core, which transfers thermal energy to the surroundings.

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